

No. of Functional Features Included in the Solution

Date	5 July 2026
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Preprocessing	Cleans missing values, encodes categorical features, and prepares data for training.	Data Preprocessing	Done	High
2	Feature Engineering	Transforms raw data into meaningful features for better model performance.	Feature Engineering	Done	High
3	Model Training	Trains Logistic Regression, Decision Tree, Random Forest, and XGBoost models.	Machine Learning	Done	High
4	Model Evaluation	Compares models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.	Model Evaluation	Done	High
5	Credit Card Approval Prediction	Predicts whether an applicant is approved or rejected.	Prediction Module	Done	High
6	Flask Web Application	Provides a user-friendly interface for entering applicant details and viewing predictions.	Frontend & Backend	Done	Medium
7	IBM Watson Deployment	Deploys the trained model on IBM Watson Machine Learning for cloud prediction.	Cloud Deployment	Done	Medium
8	Result Display	Displays approval or rejection result instantly on the web application.	User Interface	Done	Medium

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Details Form, Prediction Result Page
2	Backend / Logic	4	Data Preprocessing, Feature Engineering, Model Training, Prediction Logic
3	Database / Storage	1	Dataset Storage (CSV Files), Saved Model (.pkl)
4	API / Integration	1	IBM Watson Machine Learning Deployment
5	Security / Authentication	1	Input Validation